

COIT20258 Software Engineering

Week 7

Quality Management

Topics

- ✧ Software quality
- ✧ Software standards
- ✧ Reviews & inspections
- ✧ Quality management & agile development
- ✧ Software measurement

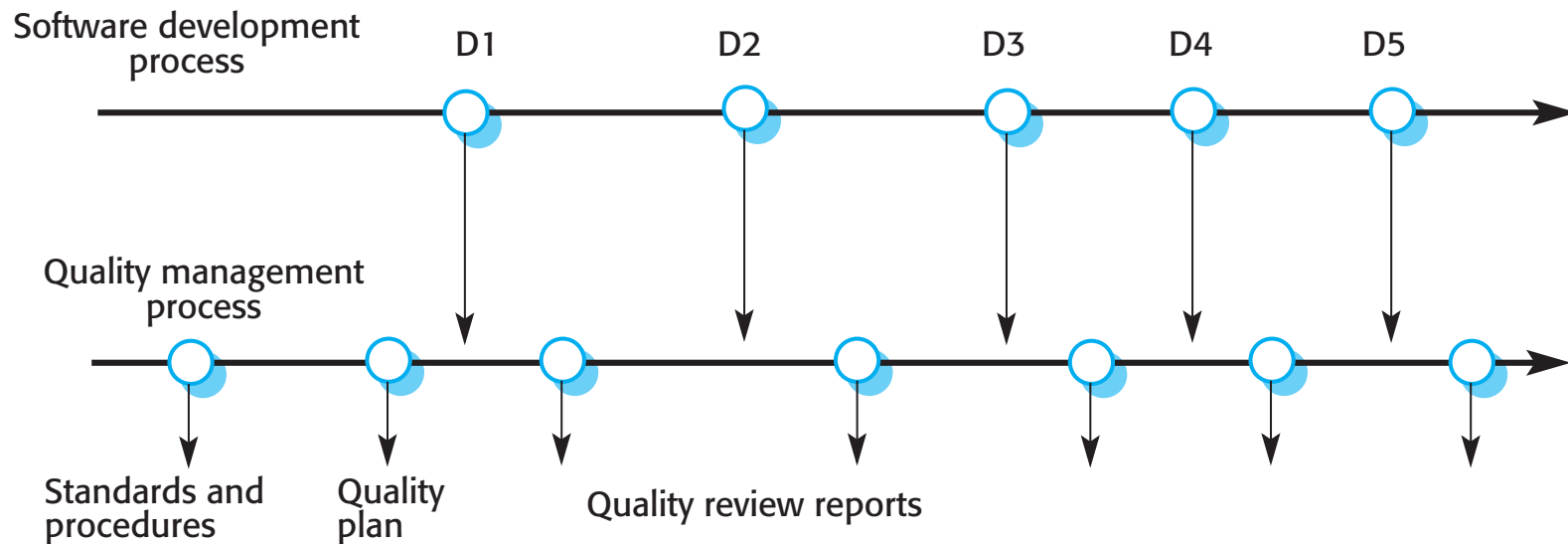
Software Quality Management

- ✧ Ensure that the required level of quality is achieved in a software product
- ✧ Three principal concerns:
 - At the organizational level, quality management is concerned with **establishing a framework** of organizational processes & standards that will lead to high-quality software
 - At the project level, quality management involves the application of specific **quality processes** & checking that these planned processes have been followed
 - At the project level, quality management is also concerned with establishing **a quality plan** for a project. The quality plan should set out the quality goals for the project & define what processes & standards are to be used.

Quality Management Activities

- ✧ Quality management provides an **independent** check on the software development process
- ✧ The quality management process checks the **project deliverables** to ensure that they are consistent with organizational standards & goals
- ✧ The quality team should be independent from the development team so that they can take an objective view of the software. This allows them to report on software quality without being influenced by software development issues.

Quality Management & Software Development



Quality Planning

- ✧ A **quality plan** sets out the desired product qualities & how these are assessed & defines the most significant quality attributes
- ✧ The quality plan should define the quality assessment process
- ✧ It should set out which organizational standards should be applied &, where necessary, define new standards to be used

Quality Plans

✧ Quality plan structure:

- Product introduction
- Product plans
- Process descriptions
- Quality goals
- Risks & risk management

✧ Quality plans should be short, succinct documents

- If they are too long, no-one will read them

Scope of Quality Management

- ✧ Quality management is particularly important for **large, complex systems**. The quality documentation is a record of progress & supports continuity of development as the development team changes.
- ✧ For smaller systems, quality management needs less documentation & should focus on establishing a quality culture
- ✧ Techniques should evolve when agile development is used

Software Quality

- ✧ Quality, simplistically, means that a product should meet its specification
- ✧ The focus may be “fitness for purpose” rather than specification conformance

Software Fitness for Purpose

- ✧ Has the software been properly tested?
- ✧ Is the software sufficiently dependable to be put into use?
- ✧ Is the performance of the software acceptable for normal use?
- ✧ Is the software usable?
- ✧ Is the software well-structured & understandable?
- ✧ Have programming & documentation standards been followed in the development process?

Non-Functional Characteristics

- ✧ The **subjective quality** of a software system is largely based on its **non-functional characteristics**
- ✧ This reflects practical user experience – if the software's functionality is not what is expected, then users will often just work around this & find other ways to do what they want to do
- ✧ However, if the software is unreliable or too slow, then it is practically impossible for them to achieve their goals

Software Quality Attributes

Safety	Understandability	Portability
Security	Testability	Usability
Reliability	Adaptability	Reusability
Resilience	Modularity	Efficiency
Robustness	Complexity	Learnability

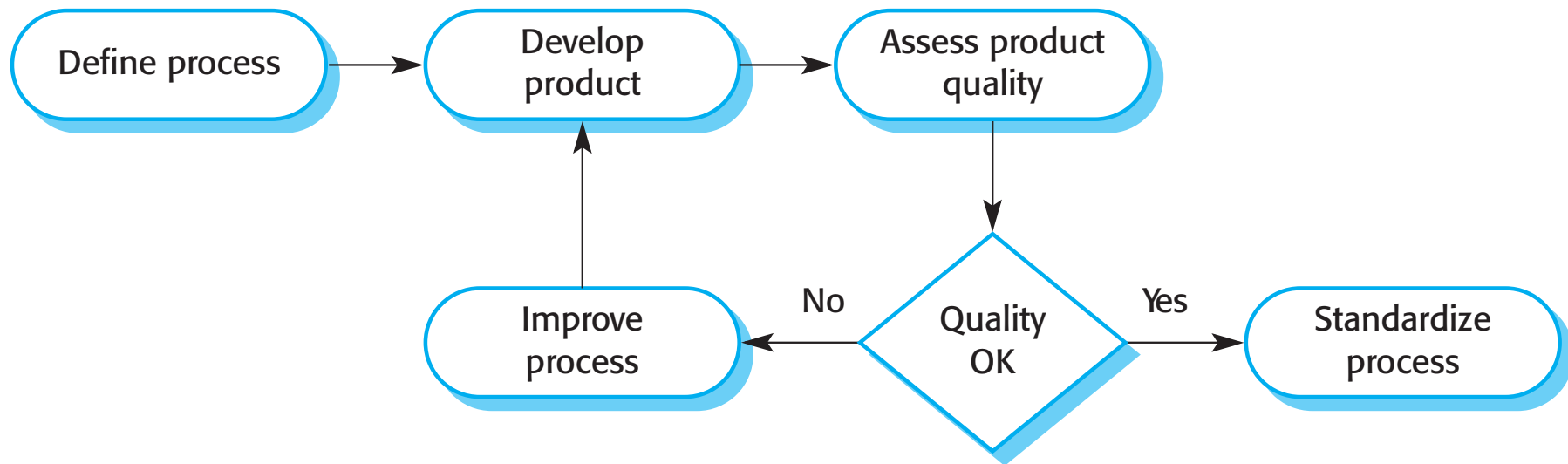
Quality Conflicts

- ✧ It is not possible for any system to be optimized for all of these attributes – for example, improving robustness may lead to loss of performance
- ✧ The quality plan should therefore define the most important quality attributes for the software that is being developed
- ✧ The plan should also include a definition of the quality assessment process, an agreed way of assessing whether some quality, such as maintainability or robustness, is present in the product

Process & Product Quality

- ✧ The quality of a developed **product** is influenced by the quality of the production **process**
- ✧ This is important in software development as some product quality attributes are hard to assess
- ✧ However, there is a very complex & poorly understood relationship between software processes & product quality
 - The application of individual skills & experience is particularly important in software development
 - External factors such as the novelty of an application or the need for an accelerated development schedule may impair product quality

Process-Based Quality



Software Standards

- ✧ Standards define the **required attributes of a product or process**. They play an important role in quality management.
- ✧ Standards may be international, national, organizational or project standards
- ✧ They are a framework for defining what quality means in a particular setting (i.e., that organization's view of quality)

Product & Process Standards

✧ Product standards

- Apply to the software product being developed. They include document standards, such as the structure of requirements documents, documentation standards, such as a standard comment header for an object class definition, & coding standards, which define how a programming language should be used.

✧ Process standards

- These define the processes that should be followed during software development. Process standards may include definitions of specification, design & validation processes, process support tools & a description of the documents that should be written during these processes.

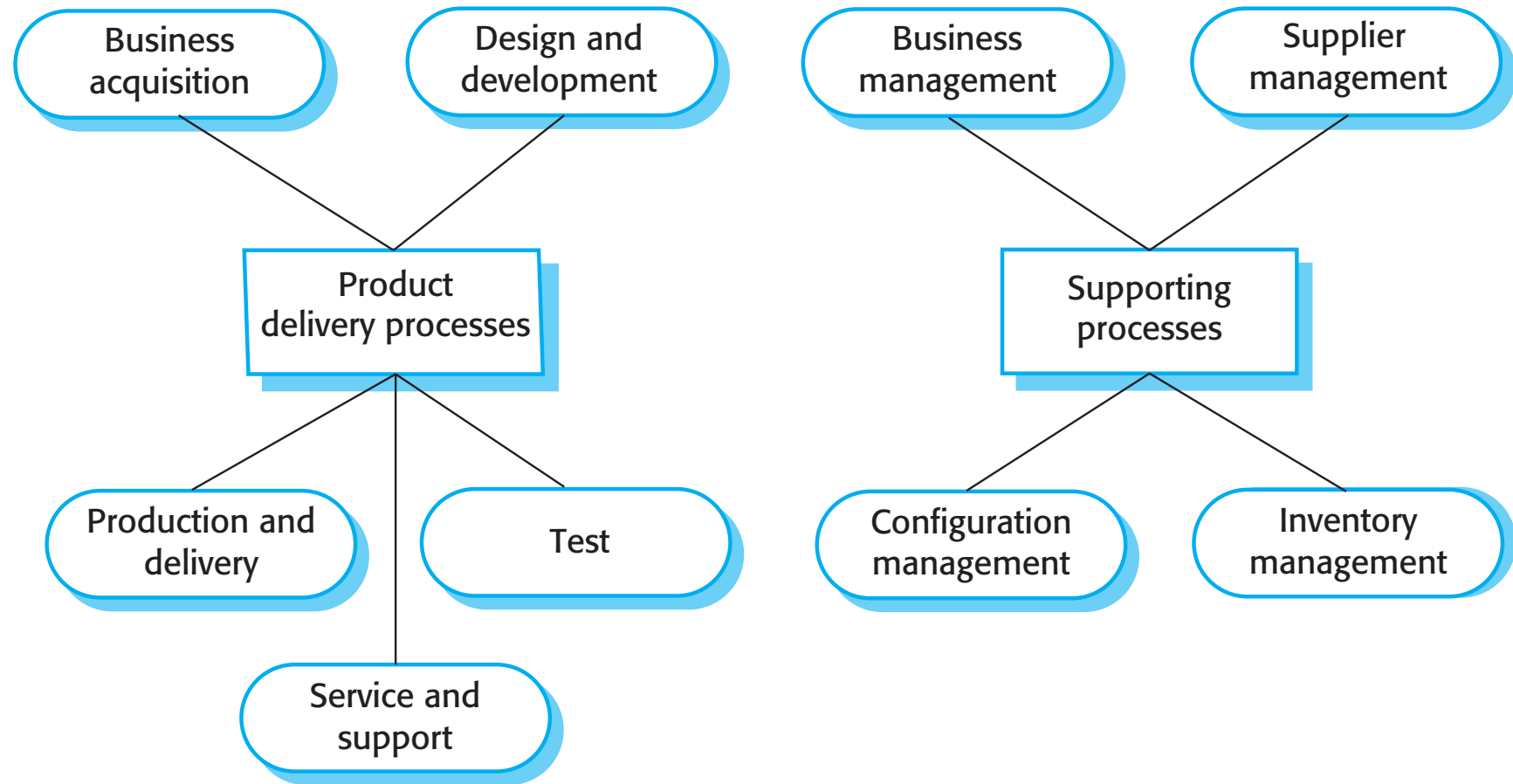
... Product & Process Standards

Product standards	Process standards
Design review form	Design review conduct
Requirements document structure	Submission of new code for system building
Method header format	Version release process
Java programming style	Project plan approval process
Project plan format	Change control process
Change request form	Test recording process

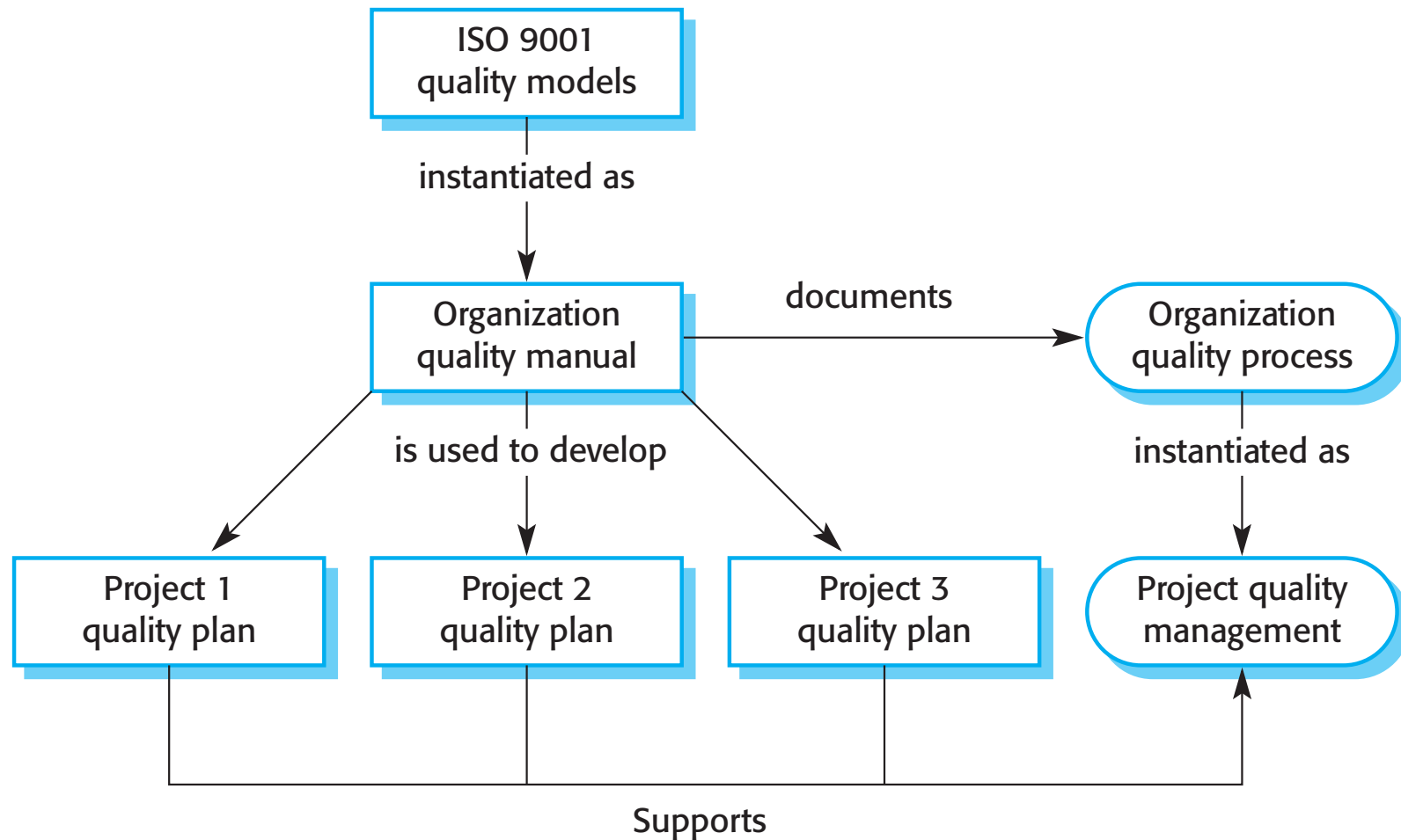
ISO 9001 Standards Framework

- ✧ An international set of standards that can be used as a basis for developing quality management systems
- ✧ **ISO 9001**, the most general of these standards, applies to organizations that design, develop & maintain products, including software
- ✧ The ISO 9001 standard is a framework for developing software standards
 - It sets out general quality principles, describes quality processes in general & lays out the organizational standards & procedures that should be defined. These should be documented in an organizational quality manual.

ISO 9001 Core Processes



ISO 9001 & Quality Management



ISO 9001 Certification

- ✧ Quality standards & procedures should be documented in an **organizational quality manual**
- ✧ An external body may certify that an organisation's quality manual conforms to ISO 9000 standards
- ✧ Some customers require suppliers to be ISO 9000 certified although the need for flexibility here is increasingly recognised

Reviews & Inspections

- ✧ A group examines part or all of a process or system & its documentation to find potential problems
- ✧ Software or documents may be “signed off” at a review which signifies that progress to the next development stage has been approved by management
- ✧ There are different types of review with different objectives
 - Inspections for defect removal (product)
 - Reviews for progress assessment (product & process)
 - Quality reviews (product & standards)

Quality Reviews

- ✧ A group of people carefully examine part or all of a software system & its associated documentation
- ✧ Code, designs, specifications, test plans, & standards can all be reviewed
- ✧ Software or documents may be “signed off” at a review which signifies that progress to the next development stage has been approved by management

Phases in the Review Process

✧ Pre-review activities

- Pre-review activities are concerned with review planning & review preparation

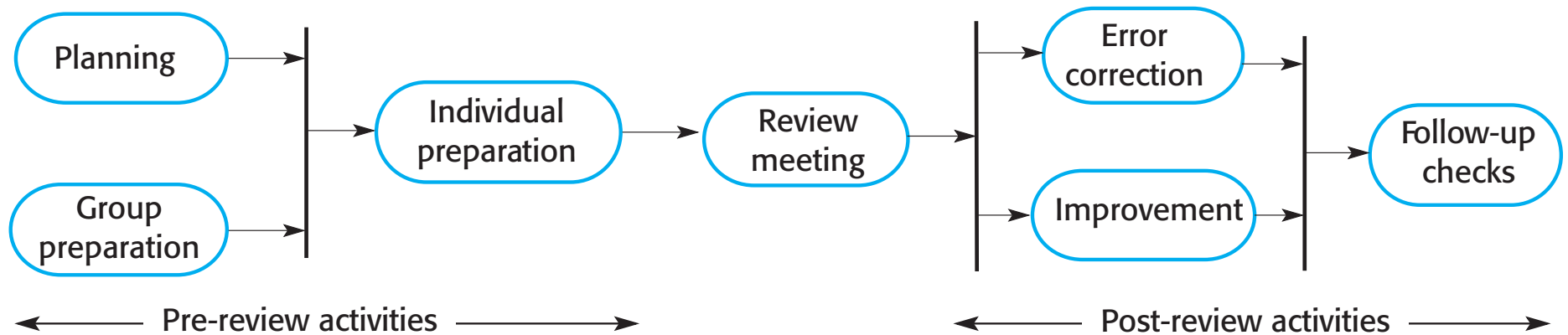
✧ The review meeting

- During the review meeting, an author of the document or program being reviewed should 'walk through' the document with the review team

✧ Post-review activities

- These address the problems & issues that have been raised during the review meeting

Software Review Process



Program Inspection

- ✧ These are peer reviews where engineers examine the source code of a system with the aim of discovering anomalies & defects
- ✧ Inspections do not require execution of a system so may be used before implementation
- ✧ They may be applied to any representation of the system (e.g., requirements, design, configuration data, test data)
- ✧ They have been shown to be an effective technique for discovering program errors

Inspection Checklist

- ✧ Checklist of common errors should be used to drive the inspection
- ✧ Error checklists are programming language dependent & reflect the characteristic errors that are likely to arise in the language

Inspection Checklist

Fault class	Inspection check
Data faults	• Are all program variables initialized before their values are used? • Have all constants been named? • Should the upper bound of arrays be equal to the size of the array or Size -1? • If character strings are used, is a delimiter explicitly assigned? • Is there any possibility of buffer overflow?
Control faults	• For each conditional statement, is the condition correct? • Is each loop certain to terminate? • Are compound statements correctly bracketed? • In case statements, are all possible cases accounted for? • If a break is required after each case in case statements, has it been included?
Input/output faults	• Are all input variables used? • Are all output variables assigned a value before they are output? • Can unexpected inputs cause corruption?

... Inspection Checklist

Fault class	Inspection check
Interface faults	• Do all function & method calls have the correct number of parameters? • Do formal & actual parameter types match? • Are the parameters in the right order? • If components access shared memory, do they have the same model of the shared memory structure?
Storage management faults	• If a linked structure is modified, have all links been correctly reassigned? • If dynamic storage is used, has space been allocated correctly? • Is space explicitly deallocated after it is no longer required?
Exception management faults	• Have all possible error conditions been taken into account?

Quality Management & Agile Development

- ✧ Quality management in agile development is **informal** rather than document-based
- ✧ It relies on establishing a **quality culture**, where all team members feel responsible for software quality & take actions to ensure that quality is maintained

Reviews & Agile Methods

- ✧ The review process in agile software development is usually informal
- ✧ In Scrum, there is a review meeting after each iteration of the software has been completed (a sprint review), where quality issues & problems may be discussed
- ✧ In Extreme Programming (XP), pair programming ensures that code is constantly being examined & reviewed by another team member

Agile QM & Large Systems

- ✧ When a large system is being developed for an external customer, agile approaches to quality management with minimal documentation may be impractical
 - If the customer is a large company, it may have its own quality management processes & may expect the software development company to report on progress in a way that is compatible with them
 - Where there are several geographically distributed teams involved in development, perhaps from different companies, then informal communications may be impractical
 - For long-lifetime systems, the team involved in development will change without documentation, new team members may find it impossible to understand development

Software Measurement

- ✧ Software measurement is concerned with deriving a numeric value for an attribute of a software product or process
- ✧ This allows for objective comparisons between techniques & processes
- ✧ Although some companies have introduced measurement programmes, most organizations still don't make systematic use of software measurement
- ✧ There are few established standards in this area

Software Metric

- ✧ Any type of measurement which relates to a software system, process or related documentation
 - Lines of code in a program, number of person-days required to develop a component
- ✧ Allow the software & the software process to be quantified
- ✧ May be used to predict product attributes or to control the software process
- ✧ Product metrics can be used for general predictions or to identify anomalous components

Types of Process Metric

✧ The time taken for a particular process to be completed

- This can be the total time devoted to the process, calendar time, the time spent on the process by particular engineers, & so on

✧ The resources required for a particular process

- Resources might include total effort in person-days, travel costs or computer resources

✧ The number of occurrences of a particular event

- Examples of events that might be monitored include the number of defects discovered during code inspection, the number of requirements changes requested, the number of bug reports in a delivered system & the average number of lines of code modified in response to a requirements change

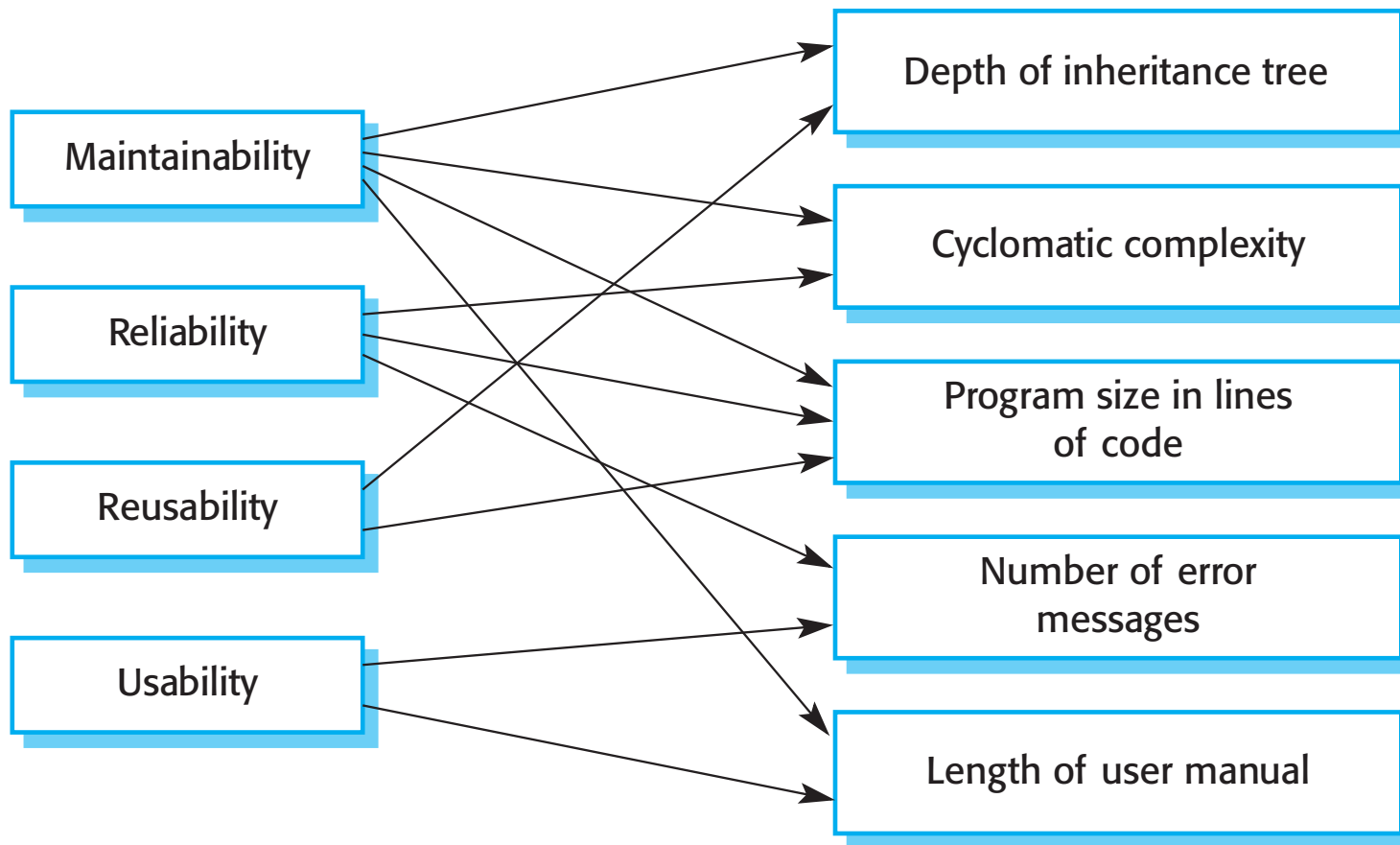
Use of Measurements

- ✧ To assign a value to system quality attributes
- ✧ To identify the system components whose quality is sub-standard

Relationships between Internal & External Software

External quality attributes

Internal attributes



Product Metrics

- ✧ A quality metric should be a **predictor of product quality**
- ✧ Classes of product metric
 - **Dynamic metrics** which are collected by measurements made of a program in execution
 - **Static metrics** which are collected by measurements made of the system representations
 - Dynamic metrics help assess efficiency & reliability
 - Static metrics help assess complexity, understandability & maintainability

Dynamic & Static Metrics

✧ **Dynamic metrics** are closely related to software quality attributes

- It is relatively easy to measure the response time of a system (performance attribute) or the number of failures (reliability attribute)

✧ **Static metrics** have an indirect relationship with quality attributes

- You need to try & derive a relationship between these metrics & properties such as complexity, understandability & maintainability

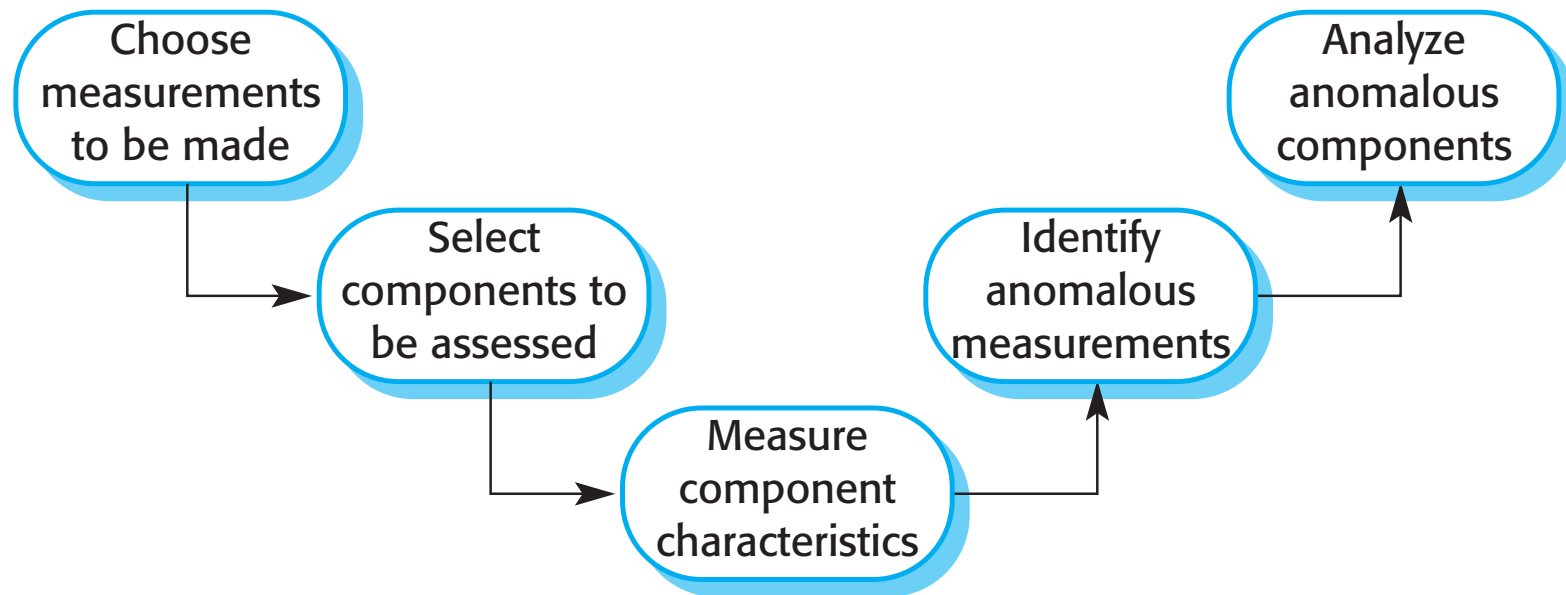
Static Software Product Metrics

Software metric	Description
Fan-in/Fan-out	Fan-in is a measure of the number of functions or methods that call another function or method (say X). Fan-out is the number of functions that are called by function X. A high value for fan-in means that X is tightly coupled to the rest of the design & changes to X will have extensive knock-on effects. A high value for fan-out suggests that the overall complexity of X may be high because of the complexity of the control logic needed to coordinate the called components.
Length of code	This is a measure of the size of a program. Generally, the larger the size of the code of a component, the more complex & error-prone that component is likely to be. Length of code has been shown to be one of the most reliable metrics for predicting error-proneness in components.

... Static Software Product Metrics

Software metric	Description
Cyclomatic complexity	This is a measure of the control complexity of a program. This control complexity may be related to program understandability. I discuss cyclomatic complexity in Chapter 8.
Length of identifiers	This is a measure of the average length of identifiers (e.g., names for variables, classes, methods) in a program. The longer the identifiers, the more likely they are to be meaningful & hence the more understandable the program.
Depth of conditional nesting	This is a measure of the depth of nesting of if-statements in a program. Deeply nested if-statements are hard to understand & potentially error-prone.
Fog index	This is a measure of the average length of words & sentences in documents. The higher the value of a document's Fog index, the more difficult the document is to understand.

Process of Product Measurement



Key Points

- ✧ Software quality management is concerned with ensuring that software has a low number of defects & that it reaches the required standards of maintainability, reliability, & portability. Software standards are important for quality assurance as they represent an identification of “best practice”. When developing software, standards provide a solid foundation for building good quality software.
- ✧ Reviews of the software process deliverables involve a team of people who check that quality standards are being followed. Reviews are the most widely used technique for assessing quality.

... Key Points

- ✧ In a program inspection or peer review, a small team systematically checks the code. They read the code in detail & look for possible errors & omissions. The problems detected are discussed at a code review meeting.
- ✧ Agile quality management relies on establishing a quality culture where the development team works together to improve software quality
- ✧ Software measurement can be used to gather quantitative data about software & the software process

... Key Points

- ✧ You may be able to use the values of the software metrics that are collected to make inferences about product & process quality
- ✧ Product quality metrics are particularly useful for highlighting anomalous components that may have quality problems. These components should then be analyzed in more detail.

*** * * End *****